

BAHAR BARGHBANI

✉ bahar.brqbni@gmail.com [in](#) [bahar barghbani](#) [github](#) [baharbarghbani](#)

Education

Sharif University of Technology

B.Sc. in Computer Engineering

September 2023 – July 2028 (Expected)

Tehran, Iran

- Overall GPA: 18.33/20.00 | Major GPA: 18.89/20.00 ([Transcript](#) [🔗](#))

Honors and Awards

National University Entrance Exam of Iran (Konkur)

2023

Tehran, Iran

- Ranked 441st among more than 136,000 participants (top 0.4%)

National Olympiad in Physics

2022

Tehran, Iran

- Bronze Medalist in National Olympiad in Physics among more than 9,000 participants.

Experience

Sabanci University

Summer 2026 – Present

Research Intern – Supervisor: Prof. Özcan Öztürk

Istanbul, Türkiye

- Developing reproducible C++ microbenchmarks for controlled characterization of processor and system performance.
- Designed a common benchmark harness with configurable iterations, warm-up execution, reproducible initialization, timing, and result validation.
- Characterizing compute, cache, and memory-system behavior across workloads.
- Building reproducible development and testing infrastructure using CMake, Git, GitHub Actions, and Linux-based measurement environments.

University of Murcia

Summer 2026 – Present

Research Intern – Supervisor: Prof. Alberto Ros

Remote

- Investigating instruction prefetching and processor performance using architectural simulation.
- Reproduced baseline performance results in gem5 and evaluated the behavior and performance impact of an instruction prefetching mechanism.
- Analyzed differences between reproduced experimental results and previously reported results by examining simulation parameters and experimental configurations.
- Currently extending the study using ChampSim for cache and instruction-prefetching experiments.

Digikala

September 2024 - November 2024

Full-Stack Developer Intern

Tehran, Iran

- Worked as a Full-Stack Developer at Digikala, the largest e-commerce company in Iran.
- Developed and maintained customer-facing pages using Next.js and React, creating reusable components and improving page load performance.
- Enhanced proficiency in React, Next.js, HTML, CSS, and Django while gaining a solid understanding of web development fundamentals.

Projects

Computer Architecture Course Project | Verilog, Logisim, ChampSim | [Github](#)

June 2024

- Designed a MIPS processor through single-cycle, multi-cycle, and pipelined implementations.
- Integrated a floating-point unit to support floating-point arithmetic operations.
- Evaluated multiple cache configurations using ChampSim and compared their effects on processor performance.

Job Shop Scheduling with Graph Neural Networks | Python, PyTorch, DGL | [Github](#)

February 2026

- Modeled Job Shop Scheduling instances as graphs and applied graph neural networks to learn operation-level scheduling decisions.
- Implemented reinforcement-learning-based training and evaluation for selecting the next schedulable operation.
- Built dataset generation, benchmarking, and evaluation utilities and compared learned scheduling policies against a random baseline.

- Developed an interactive simulator for constructing and visualizing user-defined digital logic circuits.
- Implemented real-time input manipulation and circuit evaluation with a graphical interface using PyGame.

- Implemented a simplified version-control system in C supporting commits, branching, and version tracking.
- Implemented file-management and string-processing functionality for tracking repository state and revisions.

Technical Skills

Programming Languages: C, C++, Python, Java
Computer Architecture: gem5, ChampSim, Verilog, MIPS Assembly, Logisim
Systems & Development Tools: Linux, Git, CMake, Docker, GitHub Actions
Web & Databases: Django, React, Next.js, PostgreSQL, MySQL
Documentation: LaTeX, Markdown

Coursework

Computer Architecture	Prof. Hossein Asadi	20.0/20.0
Digital System Design	Dr. Amin Foshati	20.0/20.0
Operating Systems	Prof. Hossein Asadi	19.5/20.0
Embedded Systems	Prof. Mohsen Ansari	19.6/20.0
Computer Structure and Language	Prof. Hossein Asadi	18.9/20.0
Machine Learning	Prof. Mahdieh Soleymani Baghshah	20.0/20.0
Artificial Intelligence	Dr. Mohammadreza Fereydooni	20.0/20.0
Linear Algebra	Dr. Hamid Rabiee & Dr. Maryam Ramezani	18.7/20.0
Data Structures and Algorithms	Dr. MohamadAli Abam	17.7/20.0

Voluntary and Teaching Experience

Teaching Assistant

- Fundamentals of Programming in C – Dr. Amin Fazli (Fall 2024)
- Linear Algebra – Dr. Hamid R. Rabiee (Spring 2025 & Fall 2025)
- Probability and Statistics – Dr. Amir Najafi (Fall 2025)

Voluntary Service

- Head of Sponsorship Team – Codocodile Algorithmic Contest (2025)
- Sponsorship Staff – Winter Seminar Series (WSS) 2025
- Executive Staff – ICPC 2023 Asia West Continent Final and Asian Regional in Tehran Site
- Executive Staff – ICPC 2024 Asia West Continent Final and Asian Regional in Tehran Site
- Sponsorship Staff – Codocodile Algorithmic Contest (2024)
- Sponsorship Staff – CE Intro (Introduction to Computer Engineering Event) (2024)
- Social and Content Staff – Rayan Programming Contest (2024)

Languages

English: Professional working proficiency
Persian: Native
Turkish: Intermediate proficiency
French: Beginner proficiency